CST1500: Computer Systems Architecture and Operating Systems

Adam Philpot

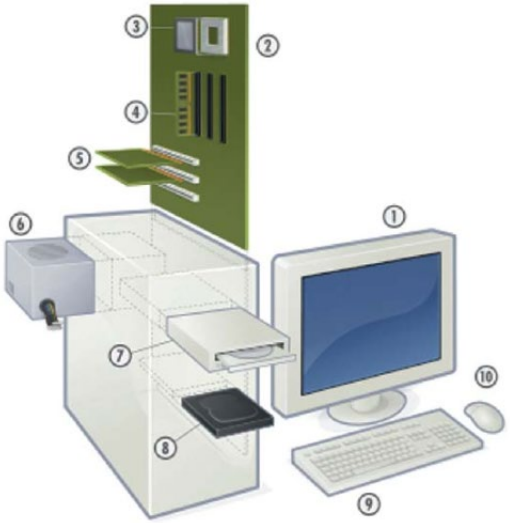
Instructors

- Dr. Adam Philpot
- Office: TG03
- Email: a.philpot@mdx.ac.uk
- Office Hours: Wednesdays 14:00 – 16:00

- Dr. Ali Khoshkholghi
- Office: T124
- Email: a.khoshkholghi@mdx.ac.uk
- Office Hours: By appointment

Goals of the module

- Understand the fundamentals of computer hardware and architecture.
- Understand the fundamentals of Operating Systems.



Hardware of a modern Personal Computer:

1. Monitor
2. Motherboard
3. CPU
4. RAM
5. Expansion cards
6. Power supply
7. Optical disc drive
8. Hard disk drive
9. Keyboard
10. Mouse

Textbooks

- Textbook for computer architecture
  - ✓ W Stallings, (2016), Computer Organization and Architecture (10th Edition), Prentice Hall. (Essential)

- Textbook for operating systems
  - ✓ A Silberschatz, P B Galvin and G Gagne, (2018), Operating System Concepts (10th Edition), ISBN 978-1-118-06333-0, John Wiley & Sons. (Essential)

Class Details – Each Week

- **3 hour lab**
  - Activities
  - Exercises
  - Tests
  - Coursework
  - Coursework Demos
- **2 hour workshop**
  - Core concepts
  - Activities
  - Code demos
  - Interaction
  - Discussion
  - Q and A
- **1 hour online session**
  - Code demos
  - Q and A
  - Student led

You MUST work outside of class to PASS the module.

Resources

- Everything you need can be found on the CST1500 course page in My Learning.
  - Weekly lab sheets
  - Lecture videos
  - Lecture slides
  - Coursework specifications
  - Coursework submission links
  - Tests
  - Further resources

Assessment

- Formative and Summative assessment.

- **Formative**: To provide feedback – does **NOT** count towards the module grade i.e. weekly lab exercises, and tests.

- **Summative**: To assess the Learning Outcomes (LO) of the module – **DOES** count towards the module grade i.e. Coursework One and Coursework Two.

Summative Assessment

- 100% coursework.

- Coursework One - 50%
  - SOB Portfolio – 10 SOBs

- Coursework Two – 50%
  - Group project – Python

- You need a total of 40% to pass the module.

Labs

- We will cover several programming languages in the lab:
  - C programming
  - Batch and Bash scripting
  - Python

- The tests will cover the material in the lectures.

- The coursework will cover all of the programming languages.

Schedule

- 12 weeks
- Each week has a lab worksheet, lectures to watch and more.
- The details are in the module handbook.
- Let’s take a look at the module handbook….

Questions